

Mitchel Carson

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AI / MACHINE LEARNING ENGINEER | ACTIVE TS/SCI

PROFESSIONAL SUMMARY

Early-career AI/ML and software engineer with U.S. Air Force experience, building dependable research and production systems. UT Austin M.S. Artificial Intelligence student with a 4.0 GPA and experience in PyTorch time-series modeling, Java/Spring Boot APIs, data pipelines, and cross-team technical delivery.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript/TypeScript, SQL, R

ML / Data: PyTorch, TensorFlow, pandas, NumPy, MLflow, Hydra, xarray

Software: Spring Boot, GraphQL, React, Node.js, PostgreSQL, Docker

Methods / Tools: Deep learning, time-series forecasting, Transformers, GRU/RNNs, temporal validation, Git, Jira

EDUCATION

University of Texas at Austin - M.S. Artificial Intelligence

Austin, TX | Expected May 2027

GPA: 4.0/4.0 | Completed: AI Ethics, Machine Learning, Deep Learning, Reinforcement Learning

Fall 2026: Advances in Deep Learning; Planning, Reasoning, and Search Under Uncertainty; Natural Language Processing

Appalachian State University - B.S. Computer Science

Boone, NC | December 2025

Cum Laude | GPA: 3.6/4.0 | Data Science Certificate | Senior Honors Thesis

RESEARCH AND SELECTED PROJECTS

HYDRA - Deep-Learning Streamflow Forecasting Research

2025 - Present

Ongoing research; manuscript in progress

- Developing a modular PyTorch residual-correction pipeline for NOAA National Water Model streamflow forecasts using NWM, USGS, and ERA5 geospatial and time-series data.
- Designed Transformer/GRU experiments, leakage-aware temporal splits, automated normalization, and hydrologic evaluation; preliminary experiments produced **26-54% lower RMSE than the LSTM baselines tested**.
- Preparing a *Water Resources Research* manuscript and maintaining versioned artifacts, diagnostics, figures, and technical reports for scientific review.

Harmony - Modular ML Software System

Active Development

- Developing a modular Python system with independently testable components for data ingestion, forecasting, validation, policy controls, and simulated execution.
- Designed time-aware evaluation, safety checks, Dockerized services, and saved run state for reproducible testing.

CONFERENCE ACTIVITIES

AGU26 Annual Meeting - Scientific Workshop Facilitator

San Francisco, CA | December 7-11, 2026

Accepted scientific workshop | HYDRA abstract under review

- Will facilitate the accepted workshop *Best Practices for AI and Agentic Workflows in Earth Science Research*, teaching earth and environmental scientists practical ways to use AI to accelerate research workflows.
- Submitted a HYDRA streamflow-forecasting abstract for the AGU26 scientific program; currently under review.

PROFESSIONAL EXPERIENCE

USAA - Software Engineering Intern

San Antonio, TX | May - August 2025

- Developed GraphQL API capabilities in Java and Spring Boot for customer-data workflows used across enterprise channels.
- Enhanced an internal troubleshooting tool by integrating source-of-record and non-source-of-record name and employment data, improving visibility for business users.
- Built JavaScript comparison views and delivered changes through an Agile/Scrum workflow using Git and Jira.
- Collaborated with product owners, engineers, and internal users to deliver usable, secure, and operationally reliable API changes.

United States Air Force - Executive Missions Aviator

JB Andrews, MD | August 2020 - April 2023

- Maintained passenger safety, logistics, and schedule reliability for distinguished guests aboard Air Force 2.
- Coordinated mission planning with flight crews, security teams, and executive staff under exacting operational standards.